NaviNote: Enabling In-Situ Spatial Annotation Authoring to Support Exploration and Navigation for Blind and Low Vision People

Summary of Supplementary Materials

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1 NOTE ON THE SUPPLEMENTARY MATERIALS

In addition to the following Supplementary Materials, this paper also has an Appendix, which can be found at the end of the full paper. The Appendix contains the participant demographics tables.

2 FORMATIVE STUDY: SCENARIO DESIGN SPACE

The scenarios we presented to participants in the Formative Study were as follows:

(1) You're in a location you know well—e.g. a park near your home—what might you want to place in the park for others (friends, family, anyone at all) to hear, or what might you want to hear from others in that location?
(2) You're at home organizing your room—e.g. organizing a bookshelf, your closet, a collection you have, etc.—would it be helpful to place audio notes in your room? What might you want to place in that space, or what might you want others to place in that space, if anything at all?
(3) You're travelling to a friend's apartment complex to house-sit for them while they're away.
    (a) Let's say either the apartment complex itself or your friend has placed notes indoors to help you find the right apartment. Would you imagine this being helpful? What kind of notes would you find helpful?
    (b) Once you find the apartment, what kind of notes might you want to hear from your friend in their apartment? These could be related to house-sitting, or not.
(4) What audio notes would you add to a space if you were telling a story to a young person (e.g. your niece/nephew/child). Where would you tell this story?
(5) What audio notes could O&M teachers add to a space to act as a training ground, and help you learn?

Author's Contact Information:

(6) How could you use audio notes to make a game? Where would you do this? Who would you make this for?
(7) You choose! Where might you place notes, or where would you want to hear notes? If you could "follow" other users of the app, who might you want to follow to hear their audio notes?

Table 1 shows our design space for the above scenarios, including whether in the scenario annotations were authored by the participant ('You') or others, whether the location was private, public, familiar or unfamiliar, and whether the activity was a specific task, for leisure, for the participant ('Yourself') or for others.

Table 1. The design space for the scenarios we presented to participants in the Formative Study. Note that since we left Scenario 7 up to the participants to choose, we leave the categories blank ('-').

	Author	**Location**		**Activity**	
Scenarios	You (Y) or Others (O)	Private (V) or Public (B)	Familiar (F) or Unfamiliar (U)	Task (T) or Leisure (L)	For Yourself (Y) or Others (O)
1	Y/O	B	F	L	Y/O
2	Y/O	V	F	T	Y
3	O	V/B	U	T/L	Y
4	Y	V/B	U/F	L	O
5	O	V/B	U/F	T	Y
6	Y	V/B	U/F	L	O
7	-	-	-	-	-

3 FORMATIVE STUDY: SEMI-STRUCTURED DISCUSSION QUESTIONS

Listing 1 shows the semi-structured discussion questions we asked during the Reflection Activity in the Formative Study.

Listing 1. The semi-structured interview questions we asked participants in the formative study.

```
1. After going outside, what were the pain points of 'using the system'?
2. What device would you want to use this on?
3. What surprised you during this workshop? Did you or others come up with any use-cases that you
thought were especially interesting/unique or you'd really like to use them for?
4. What features of the system would be most important to include?
5. What features of the system would be least important to include (or which should we definitely not
include)?
6. Anything else you want to share with us?
```

4 FORMATIVE STUDY: ANNOTATION TYPE QUOTE PERCENTAGES

There were 140 quotes from the Formative Study identified as annotations. These were categorized into six annotation types (Amenity, Safety, Accessibility, Layout, Attraction, and Experience) through thematic analysis. Some quotes were tagged multiple times (e.g., tagged with both layout and accessibility), resulting in a total tag count of 160. We present the number of tags and percent of quotes tagged with the various types in Table 2.

Table 2. The number of times quotes were tagged with annotation types reported as percent of total tags (160) and percent of total identified quotes (140).

	Amenity	**Safety**	**Accessibility**	**Layout**	**Attraction**	**Experience**
Tag Count	33	30	23	70	3	10
Percent of Tags	19.53%	17.75%	13.61%	41.42%	1.78%	5.92%
Percent of Quotes	**23.57%**	**21.43%**	**16.43%**	**50.00%**	**2.14%**	**7.14%**

5 NAVINOTE'S SYSTEM PROMPT

Listing 2 shows the prompt NaviNote uses when responding to a user query.

Listing 2. The prompt we provided our agentic system.

```
You are an AI assistant whose purpose is to aid blind and low vision user in understanding the world
around them. You are able to use a number of tools to collect and reason about information relevant to
 the user, both from sensors on their device such as GPS, compass heading, and also from the internet.
 To fulfill some requests, you may have to pipe multiple tools sequentially. Try to anticipate user
needs when possible. Never shorten the values of latitude or longitude, you should always use them at
full precision. When communicating latitude and longitude to the user, you should NEVER use the values
 directly, but always describe them as relative to the user's latitude and longitude. You should rely
on your tools to do all but the simplest of tasks, and combine the output of multiple tools together
to provide as complete as possible an answer whenever they are relevant.

From the user's perspective, you are an audio-based assistant on their phone. You hear their voice and
 respond by voice. You are a voice assistant, so disregard markdown notations like hyperlink and
asterisks for titles. You should also disregard hyperlinks per se.

Your user is visually impaired, keep that in mind and adjust your response accordingly. But don't
emphasize that to the user explicitly. You should:
- Prioritizing environmental understanding: Provide high-level overviews of space, including shape (e.
g., rectangular room, square park) and layout (object positioning relative to space, other objects,
and the user). For example, "You are in a rectangular room with a round table in the middle that's
about 2 meters wide."
- Being mindful of visual details: Ask if they want information about colors or other visual details
to personalize their experience.
- Recording visual conditions: Use tools to record details about their visual condition (e.g., totally
 blind, low vision, central vision loss) if they share it.
- Sourcing information reliably: Obtain environmental details from Layout annotations or other
reliable sources.

When you call any agents, you SHOULD ALWAYS include the User ID, which is a str(uuid.uuid4()), in your
 query. If the agent asks for it, you should call the agent again with User ID to the agent.

You have access to an agent (CustomizationAgent) that is capable of customizing the user's
requirements, including voice, directional instructions, unit preferences, visual feedback colors,
visual conditions and verbose level.

The way users interact with you is by clicking a button to start talking. They have to click the
button every time. Therefore, you should **never end your response with a question**, like "would you
like to know more about it?" as they may think they can directly speak to reply. You should instead
say something like "let me know if you want to know more" or "please tell me the specific content of
the note".
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You have access to key object details at a location (SceneGraphsAgent). Don't list numbers of key
objects in layout descriptions unless the user tells you to do so. When describing the layout, you
should choose the most important details (e.g., entrances & exits, space size, what's in the center)
and offer some position descriptions. As each user has individual language habits, you should consider
 reasonable synonyms for key objects' names. For example, table tennis table may be referred to as "
ping pong table" and gate may be referred to as "exit". However, you should still use the standard
name in the scene graph. When users ask about their location, include a specific location name if you
know. You should know two things about the user's location: latitude and longitude, and the user's x,y
,z coordinates in Unity. When it asks for latitude and longitude, you should input them; otherwise you
 should input unity coordinates. When retrieving key object positions using the user's latitude and
longitude, the key objects' positions are provided in Unity coordinates. You should not give the user
Unity coordinates directly. Instead, you should do a calculation and provide descriptions. Be aware
that, while you may not know it, the user moves all the time. Therefore, for EVERY inquiry, you should
 assume that the user has moved, and that all prior results related to a specific location aren't
valid anymore. Also, the user may refer to different instances of key objects (even the same kind) at
different times. For example, the user may ask a question about one particular podium, then move to
another one and ask "what about this one," and you should not rely on previous results to answer
questions, but rather use relevant tools each time.

You have access to an agent (BLVDBAgent) that handles all annotation-related operations. You should
hand all annoatation-relevant requests to this agent. Make sure to always pass the user ID to this
agent as this ID is necessary for most of the annotation operations.

You have access to internet search (brave search). If the question is about a key object or something
generic (i.e., not directly limited to a note or a key object), in addition to searching the notes,
you should ALSO use brave search to find answers. This can be, for example, asking for the history of
the current location or a key object (e.g., statue).

You also have access to the camera (i.e., what the user is seeing right now). Call it EVERYTIME when
needed as it frequently changes, for example, describing the scene or appearance of an object. When
the user asks about what's in front of them, you should call both the OpenAIAgent to understand the
image and key objects tool (SceneGraphsAgent). Your answer usually should combine the results,
especially include OpenAIAgent's responses. If you think key objects' information is unimportant, don'
t include it.

 You support wayfinding (SceneGraphsAgent). The user can set a destination and you will offer them
turn-by-turn instructions. You SHOULD get the position of the destination from the list of key objects
 or from an annotation. You should know the key objects' and annotations' distances from the user to
find the nearest/second nearest/furthest one. You should use tools to access the data every time and
do not rely on your previous function return. If the user wants to meet someone, you should search for
 whether that person has left an annotation somewhere, and if so, ask the user if they want to
navigate to the annotation. When the user asks about the key objects, you should be able to find out
their relative positions based on their dimensions and center positions. For example, you should know
whether a statue is inside a flower bed.

For seating, it's a two-step process: guide them to a seating area, then confirm availability. You'll
use key object locations to find seats, or help them navigate if none are nearby. Once there, use the
camera image to check for space or obstructions, guiding the user to adjust their view if needed
before confirming if the seat is free.
```

6 PRE-SET ANNOTATIONS FOR THE EVALUATION

Table 3 shows the annotations we added to the local square where participants completed the evaluation study. Note that many of the annotations were inspired by general findings about annotation types, rather than a specific quote.

Table 3. Annotations inspired by the formative workshop quotes and findings.

ID	Type	Example Quote	Annotation
1	Accessibility	If it's uneven, it'd be good to have something which tells you when there are steps and things like that.	The ground in this square is not level. It includes uneven stones and grassy areas.
2	Accessibility	So like [F2] said, [I] also like knowing where the toilets are.	There are no public toilets in this square, but there are some at the [nearby] station, including one accessible toilet.
3	Accessibility		Visitors must adhere to rules, including no ball games, cycling, or alcohol, and keeping dogs on a leash.
4	Accessibility	I think personally for me, opening times and closing times, it would have been good to know. Have a note of that, of the different buildings of the whole attraction.	[The local] square is open seven days a week, from 8am until at least 8pm in the summer.
5-6	Accessibility	You've got steps coming down. They're very wide steps or small steps coming down.	Watch out for the stairs. There are four steps. (Attached to two sets of stairs)
7	Accessibility	I would like to know if there is cracked pavement because that's something that we all struggle with and it's very unsafe for trips and falls right.	Here is a 4-meter-long ramp. Watch out for a small hole near the bottom to avoid tripping!
8	Amenities	And for me, food is really important. So where the cafes are [important].	Across the street, there's a café with arguably the best flat white in town!
9	Amenities		This notice board advertises various local community programs and support services in [the area], covering outdoor activities, health and wellbeing, and cost of living assistance.
10	Amenities	And the second thing is like a tour guide, but an audio version.	Just south, there's a block that has retail and leisure spots.
11	Amenities	[I would like to know] if there's a bus stop and what buses stop there.	The closest bus stop is at [the local toy store], just west of the square.
12	Attraction		These white and pink roses are a gift to [the city from Bulgarian locals].
13	Attraction	If we go to a place of interest, what I would expect the app to do is to make to take me to the place of attraction and to get all the details about the place of attraction.	The square has a rich history, evolving from an aristocratic residential area to a trade hub, finally becoming a public garden after WWII.
14	Attraction		There are yellow Coreopsis, pink Osteospermum and purple Salvia in the flower bed.

ID	Type	Example Quote	Annotation
15	Attraction	When I went there, it would have been nice if there was like audio notes of the history. So I had to tell fortunately there was a tour guide but then I was thinking what if he wasn't there and then he left early. I was like, no, I want to know more about the history.	This statue was accidentally won at an auction, when someone waved to greet a friend. He ended up giving the statue as a gift to [this square].
16	Attraction	The history and the context of it and all the rest.	There's confusion about whether this statue represents [a king from the early or late 1600s].
17	Attraction	It could be about nature because this is a park so you can say that it's about like tree species or something like that.	There are yellow Coreopsis, pink Osteospermum and purple Salvia in the flower bed.
18	Experience	So it would be nice to have a note, like, 'Oh, I came here with my friend, and we went to this building.'	We used the flowerbeds as a hiding spot for stolen biscuits until the ants discovered our stash.
19	Experience	It was good to reminisce about, you know, it was a nice happy childhood memory actually.	I once declared myself "King of the Square" from the podium and was dethroned five minutes later by a well-aimed pinecone.
20	Experience	I need [an annotation] because... We do a lot of work here and this is where I want them to meet me here.	Hi, this is Ben! Meet me here!
21	Experience		Hi, this is Emma! Meet me here!
22	Layout	So it's got to clearly define what someone would expect to discover when they first [arrived]. So let's say it's a building with shops either side of the main entrance.	This square has grassy areas on the outside edges and a higher platform in the center.
23	Layout	Two large fountains with pools of water close by but surrounded by thick walls.	This is the one of the two podiums with a vase on it. All the others here are bare.
24	Layout		This is one of two podiums with a vase, and the only one containing succulents.
25	Layout	There's a river beneath us... There are poles kind of next to us.	The two historical notice boards in this square are identical, talking about the history and rules.
26	Layout	Things I would like to be able to know what surround me is like if there is a zebra crossing on that street, so I can go there to cross the road safely.	There aren't any cross-walks at any of the entrances, but the streets are generally quiet.
27	Layout	Let's say you come across a structure or statue [...] describe the landmark within the park.	Two identical flower beds sit to the left and right of a central flower bed with a statue in it.

ID	Type	Example Quote	Annotation
28	Safety	It can be quite dangerous also, especially when you're on the pavement and suddenly you fall down on the side of the road.	Watch out—there's an elevated flower bed with stone edges and concave cutouts at the corners. Be careful not to trip!
29	Safety		Watch out for the podium! It's close to the flower bed's edge.
30-34	Safety		Watch out, there's a podium! (Attached to five different podiums)
35	Safety	Where the flower beds are... Please be careful not to cut your hands on the sharp thorns.	Watch out for the elevated square-shaped flower bed in the center! Be careful not to trip.
36	Safety		Watch out for the table tennis table! People love to play ping pong here.
37	Safety		Watch out for the table tennis table! Many parents bring their kids here.
38	Safety	Don't fall into the water fountain.	Watch out—there's an elevated flower bed with stone edges and concave cutouts at the corners. Be careful not to trip!
39	Safety	I am six foot two and a half and sometimes when I'm walking, even I've got my cane to guide me to avoid objects on the ground, like a tree branch hanging down, it sometimes hits me in the face. So if people who walk past that left a audio note to say, you know, tree branches are quite low on this road, and when I'm coming up to it, I was aware, then i would know when i need to duck. But I get that could change over time if every now and again the local authority cut it. But if they don't, that kind of information would be useful.	Watch out for tree branches hanging from the potted plants here!

7 EVALUATION STUDY QUESTIONNAIRE AND SEMI-STRUCTURED INTERVIEW QUESTIONS

Listing 3 shows the questionnaire from the Evaluation Study, and Listing 4 shows the semi-structured interview questions.

Listing 3. The questionnaire we asked participants to respond to after the navigation task, including subjective ratings on a 7-point Likert Scale.

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1. Can you describe the route we just navigated? [If low-vision: Please don't look back at the route,
just try your best to remember it. Experimenter can stand to block view.]
    1.1 Can you describe the route in general?
    1.2 What landmarks or objects do you recall from the route? [If they described landmarks in prev.
Q: Do you remember any other landmarks/objects]
    1.3 Considering each segment in the route, i.e. where you turned during the route,
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    1.4 Can you tell me in which direction (e.g. left or right, 3 o'clock etc. if you are facing away
from the ramp) each segment went?
    1.5 Can you give your best guess at the distance of each segment (or at least, which parts were
longer vs. shorter)?
    1.6 What landmarks or objects do you remember on each segment?
2. Did you learn anything else about the area while you were on the route?
3. On a scale of 1 to 7, where 1 means "strongly disagree" and 7 means "strongly agree."
    3.1 On a scale of 1 to 7: ``This system's capabilities meet my requirements for navigation''
        What about the system met or did not meet your requirements?
    3.2 On a scale of 1 to 7: ``This system is easy to use for navigation''
        Why do you feel this way?
    3.3 On a scale of 1 to 7: The navigation task was very mentally demanding. (Mental Demand)
        Why do you feel this way?
    3.4 On a scale of 1 to 7: The navigation task was very physically demanding. (Physical Demand)
        Why do you feel this way?
    3.5 On a scale of 1 to 7: I accomplished the navigation task I was asked to do very successfully.
(Own Performance)
        Why do you feel this way?
    3.6 On a scale of 1 to 7: I felt very insecure, discouraged, irritated, stressed, and annoyed
during this navigation task. (Frustration Level)
        Why do you feel this way?
4. Did you encounter any problems along the way? What would you like to change about the system, if
anything?
5. What did you like about the system, if anything?
6. What do you think of the amount of information? How relevant/useful was the information?
7. [after both systems] Now that you've experienced both systems, let's compare them. From what
perspectives do you find one system more effective or preferable than the other, and conversely, where
 does the other system stand out?
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Listing 4. The semi-structured interview questions we asked participants during the evalutaiton study.

```
1. What audio notes did you create, what was their purpose, and who were they for?
2. You created audio notes about this square. When outside the square, what notes would you create,
what would be their purpose, and who are they for? It could be anywhere.
3. Did you edit or delete any audio notes? How was that process?
    [If did not use the function]
    Our system also supports editing or deleting audio notes. What do you think of the functions? And,
 we use natural language for these processes, what do you think of this interaction process?
4. What are some things you learned about the area while using the system?
5. What did you think about the audio notes in the area? What was most useful, what was most
interesting, what did you want to filter out, etc.?
6. Did you filter any of the types of notes?
7. What other types of audio notes would you want, if any? [explain all six types of notes]
8. Did you customize the system in any way (e.g. voice, verbosity, colors, etc.)? If so, how?
9. What did you like best about the system?
10. What did you like least about the system- or what did you want to change or add to the system (if
anything)?
11. What do you think of the ``form factor'' of the device-a phone attached to your chest with a vest?
 Would you use something similar in real life? How might you improve it?
12. Assuming we debugged our app and it worked a lot smoother, what could you see yourself using it
for?
13. Would you come here alone and explore this square (or somewhere similar) with our app? Why or why
not?
14. Anything else you would like to share?
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8 DESIGN CONSIDERATIONS FOR INTERACTION AND FORM FACTOR

To explore how NaviNote might fit into daily routines, we also examined participants' perspectives on interaction modalities and device form factors and provide more details below.

8.1 Interaction Initiation

In our experiment, users were required to press a button each time they wanted to interact with the system. While nine participants appreciated that the press-to-speak mechanism was easy, P7, P13, and P16 suggested that a single press to initiate the back-and-forth interaction flow would be more convenient. As P13 explained, repeatedly pressing the button to ask and respond to questions felt conspicuous, and "what you don't want to do is highlight the fact that you have something [expensive]". P13 and P14 also suggested using wake words to initiate interaction with the system.

8.2 Earphones

We used bone conduction earphones for the majority of participants, with wired earphones provided as alternatives. P9, P13, and P18 appreciated that bone conduction earphones allowed them to hear both the environment and NaviNote's responses and instructions. P13 and P17 also valued their social acceptability. However, P11 and P14 reported occasional conflicts between the bone conduction earphones and their glasses, and P1, P3, and P4 chose wired earphones for personal reasons. These findings suggest that alternative earphone options should always be provided.

Fig. 1. During the study, participants wore a running vest, which held the phone with the camera facing forward. This allowed them to use NaviNote hands-free.

8.3 Front-Mounted Phone Design

We adopted a vest with Velcro straps to secure the phone in front of the user for hands-free operation in our experiment, as illustrated in Figure 1. While participants appreciated hands-free use, twelve participants raised concerns about the phone being exposed to potential thieves, and only two participants (P1, P10) mentioned considering the vest to be safer than holding the phone. Participants also reported that the vest may be cumbersome for daily use (P1, P6, P9) and socially uncomfortable (5/18). P11 and P18 noted that they sometimes unintentionally blocked the camera while the phone was held in front of them.

Four participants (P2, P4, P6, P13) suggested a phone lanyard could make wearing and taking out the phone easier. However, P4 and P17 pointed out that lanyards can shift, reducing the system's accuracy. Eight participants suggested

smaller attachable cameras instead of relying on the phone camera, whereas P6 and P12 mentioned preferring to keep all functions within a single device. P3 and P17 emphasized any additional devices should be low cost.

While P13 preferred phone-based solutions for portability, five participants recommended smart glasses, as they are hands-free (P14), socially acceptable (P7, P10), and reduce safety risks (P8, P10). In summary, there are varying opinions and trade-offs with respect to providing a hands-free solution that is safe, socially acceptable, and cost-effective.